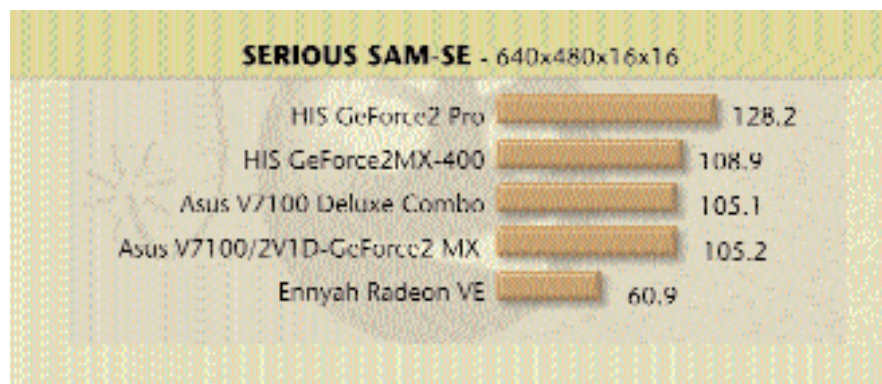




the test where some of the cards almost froze. Here, too, there was noticeable artifacting when the test was run on M64-based cards. The Trident Blade XP was very disappointing. Even though it returned an average of 25.5 fps, there was major texture loss during the major part of the test.

OpenGL

For measuring real world OpenGL gaming performance, we used Quake III Arena patched with ver 1.30 and Serious Sam: Second Edition. The Quake III engine is very unforgiving and sucks every last bit of juice from the cards when the settings are at maximum. Here the 16 MB PCI-based Matrox G450 and the Maxforce RivaTNT2 were not able to take the pressure and failed to respond when we ran the test at the maximum resolution of 1600x1200 at 32-bit colour and 32-bit texture depth. But most of the cards, including the ones above, returned very

playable scores in low-quality mode. The best score here from a GeForce2MX-200 based card was from the Asus V7100 Magic Geforce2MX-200. This card was able to churn an average of 140.3 fps, which is pretty good news considering the price. The Trident BladeXP, which came in as a surprise, scored 79.2 fps.

But the one card that shattered the fps barrier with a massive average of 355.4 fps was the HIS GeForce2 Pro. The onboard 64 MB of DDR memory is one of the reasons for such a high score. In the high-quality mode, the HIS Geforce2MX logged a score of 60.1 fps, proving that playing high-end games with some of the effects turned on would not be a problem at all.

While running the test on cards based on the MX-200 chipsets, performance was found lacking. These logged a considerably lower average of around 20 to 25 fps at high quality mode. Here, the performance loss occurs mainly because

the default high-quality settings stress the cards by turning 32-bit colour and 32-bit textures on. The Mercury Rage 128 Ultra and the Gigabyte Rage 128 Pro too just managed to crawl their way out of the benchmark with an average as low as 11.4 and 9.6 fps, respectively (at what setting?). The only card that showed its class throughout the benchmark was the HIS GeForce2 Pro with 64 MB of DDR memory. Even at an overload resolution of 1600x1200x32x32 with all the effects on, the card was able to spew out a playable average framerate of 53.3 fps.

The second OpenGL-based real world test was Serious Sam-SE, which too was conducted at three different settings. In max mode, we found that the HIS GeForce2 Pro, which blazed through the other tests, slowed down a bit and returned a score of only 29.3 fps.

Image quality

This part will include 4 screenshots with pointers. 2 of AA and 2 of AF (enabled and disabled)

The image quality test was carried out to verify the clarity of image output in



A look into the future

Present day graphics cards let you play games in 32-bit colour depth and 32-bit texture depth along with features such as Anisotropic Filtering, Environment Mapped Bump Mapping, Anti Aliasing at 4X, and Transform and Lighting. All of this can be rendered at a very playable 60 fps at a resolution of 1024x768.

But gamers demand ultimate realism and true-to-life in-game effects such as real-time shadows and near perfect facial expressions. The future 3D games based on new game engines like Doom III and Unreal II could very well incorporate these in game effects and 64-bit colour depth seems to be the way these games will take. This allows 16 bits per channel (ARGB), which translates to 65,536 levels of intensity per channel, compared to only 256 lev-

els per channel found in 32-bit colour. This would eventually mean that when multiple layers of blending, fog, haze and transparencies are used, there would be no colour banding or separation visible.

But this won't come easy; swapping textures with 64-bit colour intensity would also require higher bandwidth to facilitate smoother gameplay. This would be made possible through AGP 8X, which will soon wriggle its way onto the next generation of high-end motherboards and graphics cards; presumably it will run at 533 MHz.

Game developers would also be looking to program games that sport new visual features such as dust, realistic water textures and various other particle effects. Such high-end graphics will be made possible with the next generation of APIs such

as OpenGL 2.0 and DirectX 9.

Future games would use increasingly complex geometry which would hog much needed bandwidth. NURBS (Non Uniform Rational B Splines) seems to be the solution because it would enable complex calculations without considerably bringing down precious bandwidth. Another interesting inclusion could be Displacement Mapping. Unlike Bump Mapping which changes the surface, Displacement Mapping changes the entire object. The shadows cast by 3D textures through Bump Mapping often have a distorted look. This could get corrected by Displacement Mapping.

We could also expect features like Z-Gamma correction, higher order surfaces and real time dynamic shadowing.